## S1 Table. Data, graphs, cohorts and the four runs of the discovery-order test.

Contract-graph analysis, the main analysis of the article: approvals and transfers of WETH, USDC, USDT0, ARB and WBTC, amounts in US dollars at DefiLlama daily closing prices, every score from the twelve months before its freeze, graphs whose nodes are contracts other than smart accounts, each allowance counted up to its owner's balance at the freeze. $T_{\text{obs}}$ is 30 June 2026 and $t_1$ is 31 March 2026, each at 23:59:59 UTC. The W1 scores are the same-window scores. The two score windows overlap from July 2025 to March 2026, and the W1 score window contains the W0 label window. The discovery-order test was defined after the registered results were known; it decides no rule of the analysis plan or the registration.

### A. Events, account types, balances, graphs, universes, cohorts, labels and constants.

| | $T_{\text{obs}}$ (same window and W1) | $t_1$ (W0) |
|---|---|---|
| *Windows* | | |
| Freeze | 30 Jun 2026, 23:59:59 UTC | 31 Mar 2026, 23:59:59 UTC |
| Score window (events behind the scores) | 1 Jul 2025 – 30 Jun 2026 | 1 Apr 2025 – 31 Mar 2026 |
| Label window | 1 Jul 2026 – 30 Sep 2026 (W1) | 1 Apr 2026 – 30 Jun 2026 (W0) |
| *Events of the five tokens, 1 Apr 2025 to 30 Sep 2026 (both freezes)* | | |
| `Approval` rows of any value and non-zero `Transfer` rows | 1,240,275,042 | |
| with a contract at both ends | 603,475,784 | |
| Distinct addresses other than the zero address | 25,537,280 | |
| *Account types (code read at block 513,216,506)* | | |
| Addresses that sent a transaction (user accounts) | 17,477,779 | |
| Addresses read with `eth_getCode`[a] | 6,230,270 | |
| no code (user account) | 2,909,509 | |
| EIP-7702 delegation designator (user account) | 1,165,025 | |
| Safe (smart account, user account) | 46,204 | |
| ERC-4337 account by the rule of the reads (smart account, user account) | 87,157 | |
| ERC-4337 account by the listed code families (smart account, user account) | 192,656 (37,337 by code hash, 155,319 by code hash and implementation) | |
| any other code (contract) | 1,829,719 | |
| Typed user accounts / contracts, with the transaction senders | 21,877,279 / 1,829,719 | |
| Audited transaction senders with code / addresses with a call aborted at every attempt | 0 of 1,000 / 0 | |
| *Balances and the balance cap* | | |
| Sampled (token, owner) pairs whose balance equals `balanceOf` at the $T_{\text{obs}}$ block | 1,493 of 1,500 (99.5%; threshold 98%) | |
| Positive latest allowances between contracts | 339,282 | 308,922 |
| unlimited | 223,892 (66.0%) | 208,245 (67.4%) |
| capped to zero (the owner holds no balance) | 222,559 (65.6%) | 289,299 (93.6%) |
| capped below the allowance | 114,454 | 17,480 |
| positive after the cap | 116,723 | 19,623 |
| sum of the capped values (USD) | 68,149,271 | 83,952,622 |
| *Contract graph (1,403,959 contracts numbered over both freezes)* | | |
| Allowance layer: pairs (owner contracts / spender contracts) | 115,006 (9,459 / 100,112) | 17,940 (10,153 / 2,050) |
| pairs of the largest owner contract / of the largest spender contract | 98,073 / 2,097 | 256 / 2,008 |
| pairs without the balance cap | 302,559 | 275,917 |
| Transfer layer: pairs (sender contracts / recipient contracts) | 2,580,709 (1,007,310 / 801,813) | 2,269,073 (873,388 / 626,356) |
| pairs of the largest sender contract / of the largest recipient contract | 130,667 / 441,612 | 121,708 / 439,299 |
| Union graph: contracts with a pair / pairs | 1,299,438 / 2,575,632 | 1,111,554 / 2,269,421 |
| giant SCC / IN / OUT / other | 265,804 / 174,239 / 242,833 / 616,562 | 222,592 / 169,237 / 109,804 / 609,921 |
| *Universes, importance and hot sets of the discovery-order test* | | |
| Universe of the `both` mode (WCC of the giant SCC): contracts / union pairs | 1,169,044 / 2,378,418 | 1,015,867 / 2,121,895 |

| | $T_{\text{obs}}$ (same window and W1) | $t_1$ (W0) |
|---|---|---|
| capped allowance pairs / transfer pairs inside it | 16,776 / 2,383,503 | 17,741 / 2,121,600 |
| Universe of the `out` mode (giant SCC and OUT): contracts / union pairs | 508,637 / 1,546,906 | 332,396 / 1,283,946 |
| Typed contracts with a positive importance / with a positive $I_A$ | 809,296 / 52,446 | 715,480 / 40,395 |
| Sum of $I_T$ / of $I_A$ over the typed contracts (USD) | 16,611,212,142 / 309,472,584 | 16,622,706,250 / 343,843,976 |
| share held by the WCC: $I_T$ / $I_A$ | 96.5% / 93.0% | 97.0% / 93.8% |
| share held by the giant SCC and OUT: $I_T$ / $I_A$ | 81.3% / 90.6% | 80.7% / 87.3% |
| Hot contracts $\lvert H_q(I_T)\rvert$ / $\lvert H_q(I_A)\rvert$, `both` mode | 13,141 / 11,662 | 12,697 / 8,136 |
| Hot contracts $\lvert H_q(I_T)\rvert$ / $\lvert H_q(I_A)\rvert$, `out` mode | 5,087 / 5,088 | 3,324 / 3,324 |
| Structure pilot at $t_1$: share of the giant SCC and OUT with a capped allowance in-edge | 0.54% (stop threshold of the protocol 5%) | |
| *Cohorts (contracts) and where they lie in the contract graph of their freeze* | | |
| Cohort of the freeze ($\geq 3$ owners on the five tokens in the score window) | 6,443 of 6,483 listed (40 smart accounts are user accounts) | 6,292 of 6,343 listed (51 smart accounts); 5,351 also in the matched cohort |
| in the allowance layer / transfer layer / union graph | 1,250 (19.4%) / 4,059 (63.0%) / 4,270 (66.3%) | 1,248 (19.8%) / 3,893 (61.9%) / 4,108 (65.3%) |
| Spender cohort | 7,915 of 7,995 listed (79 user accounts, 1 system address); W1 | 7,864 of 7,962 listed (97 user accounts, 1 system address); W0 |
| in the allowance layer / transfer layer / union graph | 1,975 (25.0%) / 5,504 (69.5%) / 5,759 (72.8%) | 1,901 (24.2%) / 5,443 (69.2%) / 5,700 (72.5%) |
| outside the contract graph (score zero) | 2,156 (27.2%) | 2,164 (27.5%) |
| *Labels of the spender cohorts (they count every account, user accounts included)* | | |
| New approval pairs: total (share from user accounts); spenders with the label | 702,275 (87.9%); 1,491 | 748,258 (90.4%); 1,725 |
| New transfer senders: total (share from user accounts); spenders with the label | 618,461 (82.3%); 1,369 | 651,474 (86.3%); 1,416 |
| Revocations: total (share from user accounts); spenders with the label | 43,625 (96.9%); 935 | 61,869 (97.2%); 1,113 |
| Drained owners (heuristic): total (share from user accounts); spenders with the label | 528,557 (95.3%); 1,611 | 570,724 (95.9%); 1,807 |
| *Constants (USD, from the W0 score window of the contract graph, two significant digits)* | | |
| Median transfer $m_T$ / median positive capped allowance $m_A$ | 160 / 0.071 | |
| Cap $U$ of the amount-weighted allowance layer / cap $U_{\text{user}}$ per user account in the importance | 16,000 / 31,000 | |

A cell that spans both columns holds for both freezes. SCC: giant strongly connected component of the union graph; IN: contracts that reach it; OUT: contracts reached from it; WCC: the weakly connected component that holds the giant SCC. A user account is an address that sent a transaction, an address without code, an address with an EIP-7702 delegation designator or a smart account; every other typed address is a contract. $I_T$: US dollars that user accounts transferred to a contract in the score window; $I_A$: US dollars that they had authorized it to spend at the freeze, counted up to their balance; both counted up to $U_{\text{user}}$ per account. $H_q(I)$: the contracts of a universe whose importance $I$ is positive and among its largest, $q = 1\%$ of the universe with ties. A cohort list holds the addresses with contract code that received a non-zero approval of one of the five tokens from at least three distinct owners in the score window; a spender list holds the addresses with a positive latest allowance of one of the five tokens, set in the score window, at the freeze, among the approvals with an address of the cohort list as owner or spender. [a]The addresses that sent no transaction and have a counterparty that is neither a transaction sender nor the zero address, the members of the cohort and spender lists, and two audit samples of 1,000 addresses each.

**B.** $\mathrm{AUC}_{10\%}$ **of every ordering in the four runs.**

| | $T_{\mathrm{obs}}$, `both` (decides) | | $T_{\mathrm{obs}}$, `out` | | $t_1$, `both` | | $t_1$, `out` | |
|---|---|---|---|---|---|---|---|---|
| Ordering | $H_q(I_T)$ | $H_q(I_A)$ | $H_q(I_T)$ | $H_q(I_A)$ | $H_q(I_T)$ | $H_q(I_A)$ | $H_q(I_T)$ | $H_q(I_A)$ |
| Breadth-first (BFS) | 0.0565 | 0.0882 | 0.1179 | 0.0595 | 0.0543 | 0.1088 | 0.1269 | 0.0595 |
| Random | 0.0609 | 0.0725 | 0.1085 | 0.0621 | 0.0543 | 0.0930 | 0.1169 | 0.0703 |
| $\mathrm{IB}'_A$ (allowance backlinks) | 0.0626 | 0.0881 | 0.1438 | 0.0818 | 0.0539 | 0.1094 | 0.1750 | 0.1076 |
| $\mathrm{IB}'_T$ (transfer backlinks) | 0.1135 | 0.1466 | 0.3023 | 0.1233 | 0.1107 | 0.2001 | 0.3460 | 0.1478 |
| $\mathrm{IB}'_U$ (contract backlinks) | 0.1136 | 0.1514 | 0.3026 | 0.1302 | 0.1276 | 0.2328 | 0.3461 | 0.1561 |
| EndorseRank′ | 0.0621 | 0.0869 | 0.1404 | 0.0802 | 0.0526 | 0.1062 | 0.1733 | 0.1068 |
| EndorseRank-activity′ | 0.0612 | 0.1042 | 0.1404 | 0.0802 | 0.0587 | 0.1426 | 0.1734 | 0.1066 |
| AWP′ | 0.1465 | 0.1582 | 0.4446 | 0.1753 | 0.1524 | 0.2220 | 0.4766 | 0.2231 |
| C-PR′ ($\lambda = 1$) | 0.0617 | 0.0864 | 0.1398 | 0.0789 | 0.0526 | 0.1057 | 0.1728 | 0.1063 |
| C-PR′ ($\lambda = 0$) | 0.1061 | 0.0806 | 0.4133 | 0.1584 | 0.1088 | 0.1097 | 0.4818 | 0.2020 |
| C-PR′ ($\lambda = 0.25$) | 0.1081 | 0.0930 | 0.4129 | 0.1835 | 0.1110 | 0.1297 | 0.4817 | 0.2349 |
| C-PR′ ($\lambda = 0.5$) | 0.1085 | 0.0943 | 0.3829 | 0.1738 | 0.1118 | 0.1319 | 0.4683 | 0.2300 |
| C-PR′ ($\lambda = 0.75$) | 0.1091 | 0.0952 | 0.3744 | 0.1608 | 0.1121 | 0.1323 | 0.4562 | 0.2197 |
| S-PR′ | 0.1069 | 0.0861 | 0.4190 | 0.1805 | 0.1095 | 0.1170 | 0.4874 | 0.2360 |
| $\delta'$ (diagnostic, outside every comparison) | 0.1643 | 0.1942 | 0.5357 | 0.2317 | 0.1775 | 0.2778 | 0.5795 | 0.2937 |
| Reference: order by C-PR of the full graph | 0.1061 | 0.1157 | 0.2449 | 0.1279 | 0.1165 | 0.1646 | 0.3239 | 0.1773 |
| Reference: order by union in-degree of the full graph | 0.1593 | 0.1994 | 0.4600 | 0.2142 | 0.1757 | 0.3151 | 0.5826 | 0.2846 |
| Reference: ideal order | 0.9438 | 0.9501 | 0.9500 | 0.9500 | 0.9375 | 0.9600 | 0.9500 | 0.9500 |
| Reference: random curve | 0.0500 | 0.0500 | 0.0500 | 0.0500 | 0.0500 | 0.0500 | 0.0500 | 0.0500 |
| Contracts of the universe | 1,169,044 | | 508,637 | | 1,015,867 | | 332,396 | |
| Hot contracts | 13,141 | 11,662 | 5,087 | 5,088 | 12,697 | 8,136 | 3,324 | 3,324 |

$\mathrm{AUC}_{10\%}$ is the mean over the hot contracts of $\max(0, 1 - t_h/\lceil 0.1|\mathcal{U}|\rceil)$, with $t_h$ the visit at which a crawl reaches hot contract $h$ and $\mathcal{U}$ the universe of the run, averaged over 20 seeds. In the `both` mode a visit reveals a contract's in-edges and out-edges and the universe is the WCC; in the `out` mode a visit reveals out-edges only and the universe is the giant SCC with OUT. A primed name is the count or walk computed on the contracts read so far; $\mathrm{IB}'_A$, $\mathrm{IB}'_T$ and $\mathrm{IB}'_U$ count the visited contracts with an allowance edge, a transfer edge or an edge of either layer to the frontier contract. C-PR′ ($\lambda = 1$) and C-PR′ ($\lambda = 0$) are the allowance and the transfer layer of the coupled walk walked alone. The references are not crawls. The decision of the test is read in the first two columns only (Table 2 of the article, with intervals).

## C. Contrasts outside the decision in the three runs that carry no verdict (items 1 and 2 of the protocol's list, and the replication contrast).

| | $T_{\text{obs}}$, `out` | | $t_1$, `both` | | $t_1$, `out` | |
|---|---|---|---|---|---|---|
| Contrast | On $H_q(I_T)$ | On $H_q(I_A)$ | On $H_q(I_T)$ | On $H_q(I_A)$ | On $H_q(I_T)$ | On $H_q(I_A)$ |
| *1. Each walk′ against the count it smooths* | | | | | | |
| EndorseRank′ − $\text{IB}'_A$ | −0.0033 [−0.0038, −0.0029] | −0.0016 [−0.0019, −0.0013] | −0.0012 [−0.0014, −0.0011] | −0.0032 [−0.0034, −0.0030] | −0.0017 [−0.0020, −0.0014] | −0.0008 [−0.0011, −0.0006] |
| EndorseRank-activity′ − $\text{IB}'_A$ | −0.0033 [−0.0038, −0.0029] | −0.0016 [−0.0019, −0.0012] | +0.0049 [+0.0035, +0.0061] | +0.0332 [+0.0300, +0.0362] | −0.0016 [−0.0020, −0.0013] | −0.0010 [−0.0012, −0.0008] |
| C-PR′ ($\lambda = 1$) − $\text{IB}'_A$ | −0.0040 [−0.0044, −0.0036] | −0.0029 [−0.0032, −0.0026] | −0.0013 [−0.0014, −0.0012] | −0.0037 [−0.0039, −0.0035] | −0.0022 [−0.0025, −0.0019] | −0.0013 [−0.0015, −0.0010] |
| AWP′ − $\text{IB}'_T$ | +0.1423 [+0.1289, +0.1543] | +0.0520 [+0.0424, +0.0611] | +0.0417 [+0.0381, +0.0452] | +0.0219 [+0.0157, +0.0284] | +0.1307 [+0.1167, +0.1456] | +0.0752 [+0.0624, +0.0870] |
| C-PR′ ($\lambda = 0$) − $\text{IB}'_T$ | +0.1110 [+0.0992, +0.1229] | +0.0351 [+0.0258, +0.0431] | −0.0019 [−0.0056, +0.0017] | −0.0904 [−0.0971, −0.0838] | +0.1358 [+0.1216, +0.1504] | +0.0542 [+0.0418, +0.0657] |
| C-PR′ − $\text{IB}'_A$ | +0.2391 [+0.2283, +0.2486] | +0.0920 [+0.0837, +0.1003] | +0.0579 [+0.0538, +0.0616] | +0.0224 [+0.0165, +0.0281] | +0.2933 [+0.2799, +0.3052] | +0.1224 [+0.1103, +0.1331] |
| C-PR′ − $\text{IB}'_T$ | +0.0806 [+0.0692, +0.0922] | +0.0504 [+0.0408, +0.0596] | +0.0010 [−0.0025, +0.0046] | −0.0683 [−0.0756, −0.0609] | +0.1223 [+0.1078, +0.1363] | +0.0822 [+0.0682, +0.0937] |
| *2. Among the walks* | | | | | | |
| C-PR′ − C-PR′ ($\lambda = 0$) | −0.0305 [−0.0343, −0.0264] | +0.0153 [+0.0108, +0.0206] | +0.0030 [+0.0021, +0.0038] | +0.0222 [+0.0195, +0.0254] | −0.0135 [−0.0182, −0.0090] | +0.0280 [+0.0213, +0.0350] |
| C-PR′ − AWP′ | −0.0617 [−0.0709, −0.0526] | −0.0016 [−0.0091, +0.0063] | −0.0406 [−0.0442, −0.0371] | −0.0901 [−0.0967, −0.0832] | −0.0084 [−0.0182, +0.0028] | +0.0070 [−0.0029, +0.0167] |
| C-PR′ − C-PR′ ($\lambda = 1$) | +0.2431 [+0.2316, +0.2525] | +0.0949 [+0.0865, +0.1033] | +0.0592 [+0.0551, +0.0629] | +0.0262 [+0.0202, +0.0317] | +0.2955 [+0.2820, +0.3074] | +0.1237 [+0.1116, +0.1343] |
| C-PR′ − EndorseRank′ | +0.2424 [+0.2312, +0.2518] | +0.0936 [+0.0852, +0.1020] | +0.0592 [+0.0551, +0.0628] | +0.0256 [+0.0197, +0.0312] | +0.2950 [+0.2816, +0.3070] | +0.1233 [+0.1112, +0.1340] |
| S-PR′ − C-PR′ ($\lambda = 0$) | +0.0057 [+0.0041, +0.0078] | +0.0221 [+0.0185, +0.0256] | +0.0006 [+0.0005, +0.0009] | +0.0073 [+0.0059, +0.0090] | +0.0056 [+0.0034, +0.0078] | +0.0341 [+0.0283, +0.0398] |
| *4. Replication contrast: C-PR′ − $\text{IB}'_U$ on the hot set of the union in-degree* | | | | | | |
| C-PR′ − $\text{IB}'_U$ on $H_q(\text{IB}_U)$ | −0.1955 [−0.2086, −0.1819] | | −0.4689 [−0.4762, −0.4617] | | −0.0877 [−0.1017, −0.0724] | |

Differences of $\text{AUC}_{10\%}$ between two orderings with the 95% percentile interval of a paired bootstrap over the hot contracts (400 resamples, seed 42); an interval above 0 means that the first ordering reached the hot contracts sooner. C-PR′ without a value of $\lambda$ is C-PR′ at $\lambda = 0.5$. $H_q(\text{IB}_U)$ is the hot set of the union in-degree of the full graph. The same contrasts of the deciding run are Table 3 of the article.

**D. Descriptive readings of the deciding run ($T_{\text{obs}}$, `both` mode), by ordering.**

| | Hot contracts visited after 10% of the visits | | Importance held by the contracts visited, after 1% | | Importance held by the contracts visited, after 10% | | Picks decided by a tie | | Wasted visits in the first | | Per crawl (mean over seeds) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ordering | $H_q(I_T)$ | $H_q(I_A)$ | $I_T$ | $I_A$ | $I_T$ | $I_A$ | `both` | `out` | 1% | 10% | Solves | Iterations |
| Breadth-first (BFS) | 10.9% | 17.8% | 7.6% | 6.0% | 56.9% | 53.5% | 0.000 | 0.000 | 0.891 | 0.845 | 0 | 0 |
| Random | 10.0% | 10.0% | 6.5% | 5.2% | 41.4% | 36.9% | 0.000 | 0.000 | 0.841 | 0.682 | 0 | 0 |
| $\text{IB}'_A$ (allowance backlinks) | 11.0% | 15.3% | 13.1% | 34.5% | 64.1% | 68.4% | 1.000 | 1.000 | 0.874 | 0.862 | 0 | 0 |
| $\text{IB}'_T$ (transfer backlinks) | 16.1% | 23.9% | 42.1% | 25.9% | 74.3% | 50.1% | 1.000 | 0.999 | 0.678 | 0.888 | 0 | 0 |
| $\text{IB}'_U$ (contract backlinks) | 16.1% | 24.7% | 42.1% | 47.7% | 74.3% | 85.9% | 1.000 | 0.999 | 0.675 | 0.888 | 0 | 0 |
| EndorseRank′ | 11.0% | 15.2% | 13.1% | 34.5% | 64.0% | 68.4% | 0.999 | 0.998 | 0.874 | 0.863 | 540 | 1,322 |
| EndorseRank-activity′ | 9.4% | 15.4% | 28.7% | 51.4% | 64.4% | 86.5% | 0.991 | 0.997 | 0.692 | 0.857 | 540 | 13,805 |
| AWP′ | 17.6% | 21.2% | 80.0% | 46.4% | 82.0% | 49.2% | 0.252 | 0.005 | 0.522 | 0.835 | 542 | 29,042 |
| C-PR′ ($\lambda = 1$) | 10.9% | 15.2% | 13.1% | 34.5% | 64.1% | 68.4% | 1.000 | 0.999 | 0.874 | 0.863 | 540 | 1,317 |
| C-PR′ ($\lambda = 0$) | 14.0% | 12.0% | 70.7% | 45.0% | 73.1% | 49.7% | 0.531 | 0.019 | 0.747 | 0.911 | 540 | 24,303 |
| C-PR′ ($\lambda = 0.25$) | 14.1% | 13.2% | 71.0% | 82.9% | 73.3% | 86.9% | 0.534 | 0.019 | 0.727 | 0.908 | 540 | 23,328 |
| C-PR′ ($\lambda = 0.5$) | 14.4% | 13.7% | 71.4% | 84.1% | 73.5% | 87.0% | 0.535 | 0.023 | 0.719 | 0.900 | 540 | 22,594 |
| C-PR′ ($\lambda = 0.75$) | 14.6% | 13.8% | 71.4% | 84.1% | 73.5% | 87.0% | 0.537 | 0.025 | 0.719 | 0.889 | 540 | 22,273 |
| S-PR′ | 14.0% | 12.8% | 70.7% | 76.7% | 73.3% | 86.4% | 0.531 | 0.019 | 0.743 | 0.910 | 1,080 | 26,024 |
| $\delta'$ (diagnostic) | 18.9% | 27.3% | 73.6% | 87.6% | 77.7% | 88.9% | 0.634 | 0.283 | 0.497 | 0.650 | 0 | 0 |

Hot contracts visited: the discovery curve, the share of the hot set that a crawl has visited, read at the grid point nearest 10% of the universe (mean over the 20 seeds); the random curve is at 10% there and the ideal order at 100%. Importance held: the share of the summed importance of the universe that the visited contracts hold, read at the grid points nearest 1% and 10% of the universe (mean over the seeds); the protocol reports these curves without intervals. Picks decided by a tie: the share of picks whose group of tied scores held more than one contract, in the deciding run and in the `out` mode at $T_{\text{obs}}$. Wasted visits: the share of the first 1% and of the first 10% of the visits that went to contracts with $I_T = I_A = 0$. Solves and power iterations: per crawl at the schedule of the deciding run ($\rho = 1/50$, about 540 recomputes of the ordering per crawl); S-PR′ solves two walks at each recompute. None of these readings enters a verdict.